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Past-Papers-2016

Short

what is supervised learning?

what is unsupervised learning?

what is knowledge?

what is support in data mining?

what do you mean by confidence in data mining?

what is posterior probability?

what is entropy?

what do you mean by nominal data?

what is the time consumed by k-mean algorithm?

$$O(\overset{\text{iterations}}{I} \times \underbrace{N}_{\text{no of data points}} \times \underbrace{K}_{\text{number of clusters}} \times \underbrace{D}_{\text{dimensionality of data}})$$

Give example of ordinal data?

- Economic status (lower class, upper class, higher class)
- Rating scale
- Likert scales (agree, disagree)

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what is descriptive data mining?

⇒ main goal is to summarize & explore or Turn given data into relevant information.

⇒ Identify patterns and r/s in data.

⇒ Data profilling

Types

- Clustering Analysis
- Summarization
- Association
- sequence discovery Analysis

Name any technique that is more efficient than aprori?

- Frequent Pattern Growth method.

Discuss disadvantages of k-mean algorithm?

- Sensitive to outliers
- Sensitive to initial centroid selection

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- Assume spherical clusters of equal size
- Require predefined number of clusters
- may converge to local optima
- Not suitable for non-linear data.
- Does not incorporate noise handling.

Q) How do you find polarity for nominal data?

⇒ Polarity refer to +ive or -ive sentiment associated with textual or categorical data.

$$\text{polarity}(+\text{ive}) = \frac{\text{no of +ive instances}}{\text{no of instances in category}} \times 100$$

methods:

⇒ frequency based polarity

⇒ Domain Specific knowledge

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Give example of ordinal data

what do you mean by maximal in maximal frequent pattern growth mining?

⇒ A maximal pattern is a set of items that occur frequently in the data, and there is no larger set of items with the same frequency

⇒ it's "maximal" because it cannot be expanded further without losing its frequency property.

How is k-medoids algorithm is better than k-means?

- Robust to outliers
- can handle clusters of unequal size
- work with arbitrary distance metrics.
- Handle non-numeric data.
- Scalability
- Deterministic results

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Give an example of noise and inconsistent data?

- Noise can distort patterns & r/s in data.
- Inconsistent can lead to error and inaccuracies in analysis outcome

Ex: a score of 1000 out of 100, is outlier or data entry error.

inconsistent: existing -ive value in age or > 150 years.

what happen if we train the machine with false data?

- can mess up learning process
- wastage of resources & time.
- making biased predictions/classification
- leading to wrong decisions.
- Poor performance

⇒ To avoid these problem:

- clean data
- accurate data
- Testing the model

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M T W T F S

Is privacy of data an issue in data mining? Give reason?

⇒ Yes, privacy of data is significant issue in data mining.

- can lead to identity theft or discrimination.
- Data breaches
- inference attack

⇒ To avoid this:

- implement data protection measures
- Privacy-preserving data mining
- methods.

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